Final Network Exam
Fall 2023
Name:

90 minutes

Q1 (5 points)- Compare the RSA, and DES algorithms.

Q2 (5 points)-What is the purpose of nonce in authentication protocols?

Q3 (5 points)-What is the role of certificate authority?

Q4 (5 points)-Explain how to receive an email securely (confident, integrity and authentication)

Your answer must include visual illustration for your answer.

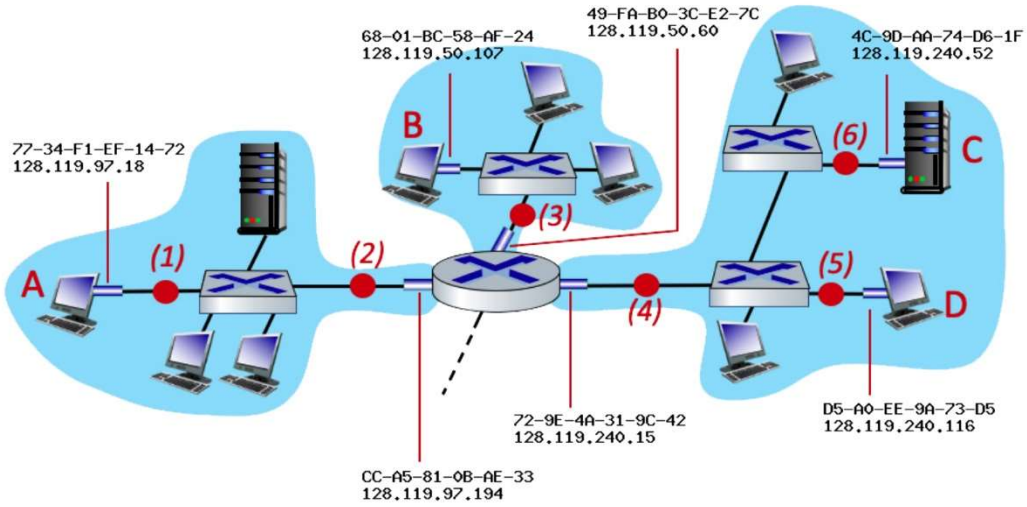
Q5 (10 points) Multiple Access protocols. Assume a multiple access wireless network, where 6 messages have been submitted from different nodes at times t = 0.8, 1.5, 2.0, 2.1, 2.7, 3.1.

- Each transmission requires exactly one time unit.
- You can assume that if a packet experiences a collision, a node will not attempt a retransmission of that packet until sometime after t=6.
- You should assume that it takes 0.2 time units for a signal to propagate from one node to each of the other nodes.

For each of the following protocols indicate which packets are successfully transmitted.

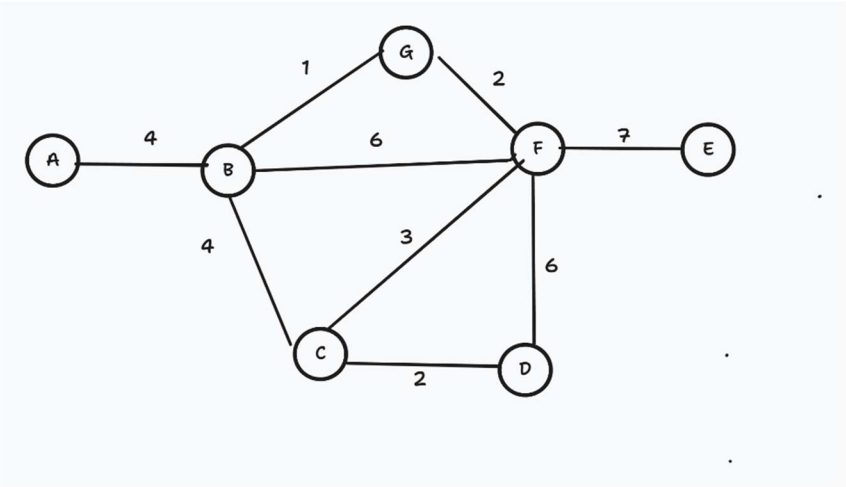
a) Pure ALOHA
b) Slotted ALOHA
c) CSMA protocol (**without** collision detection)
d) CSMA/CD protocol (**with** collision detection)

Q6 (10 points) Consider the network shown below. The IP and MAC addresses are shown for hosts A, B, C and D, as well as for the router's interfaces.



Assume IP Datagram from B-> C			
Src IP at 3	Dest IP at 3	Src MAC at 3	Dst MAC at 3
Src IP at 4	Dest IP at 4	Src MAC at 4	Dst MAC at 4
Src IP at 6	Dest IP at 6	Src MAC at 6	Dst MAC at 6

Q7 (10 points) Dijkstra's Algorithm Consider the network shown below, and Dijkstra's link-state algorithm to find the least cost path from source node to all other destinations. Compute the shortest path starting from router A and the forwarding table at router A



Q8 (10 points) Compare between traditional, and software defined networking routing.

Explain how SDN can implement a switch, router and firewall?

Q9 (5 points) Discuss the difference within Autonomous System (AS) and across Autonomous System (AS). (i.e. Intra-AS,) Highlight protocols and the goal of routing in each scenario.

Q10 (5 points) Why does routing use IP addresses but not MAC addresses?

Q11 (2 points) Compare circuit vs packet switching?

Q12 (2 Points) in BGN, the sender windows is [100,120], what happened when receive an acknowledge for 110?

Q13 (2 Points) What is piggy backing in TCP, and why it is used?

Q14 (2 Points) Explain the role of rwnd in TCP header.

Q15 (4 points) A TCP connection has a Receiver Window of 18KB. The congestion window grows as shown in the table below:

1. What is the value of the slow-start threshold at the beginning of the transmission?
2. What is the new value of slow-start threshold?
3. What is the cwnd at time 12 and 13.

Transmission Number	1	2	3	4	5	6	7	8	9	10	11	12	13
cwnd	1	2	4	8	16	17	18	19	1	2	4	..	.

Q16 (4 points) Suppose a client is downloading a 8 MB file from a Server, through a single communication link, with bandwidth capacity of 100 Mbps. If the distance between the Client and the Server is 10,000 Km and the propagation speed through the medium is $2 * 10^8$ m/s.

Compute transmission and the propagation delay.

Q 17 (4 points) Why does the header checksum change at each router in IPv4?